

Harshil Patel

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EDUCATION

Durham University, BSc Computer Science

On track for 1st class

Durham, England

Sep. 2025 – Exp. Jun 2028

EXPERIENCE

Snapchat

Jul 2026 – Aug 2026

Augmented Reality Development Intern | JavaScript, LLM Integration

London, England

- Owned end-to-end development of an AR Horoscope Lens from prototype to production, shipping a customer-facing feature that reached **50,000+ views**
- Architected an event-driven JavaScript runtime across 5 modules, coordinating real-time animations and face-tracked rendering in a production system
- Integrated the OpenAI API with structured JSON prompting to drive real-time, personalised LLM-generated content – early hands-on experience shipping an LLM-backed feature to production users

Data Engineering Research Intern

June 2026 – August 2026

Department of Engineering, Durham University | Python, Pandas

Durham, England

- Engineered a Python/Pandas ETL pipeline unifying 10+ datasets into a single GDPR-compliant data store, with automated data quality gates and end-to-end lineage logging
- Identified and corrected two systematic data quality faults in the source datasets, affecting **1,300+ student records** and **300 schools**
- Designed a rules-based eligibility model spanning 11 research questions, translating complex admissions policy into an actionable, documented data model

Expedia Group

Jul 2026

Spring Week – Technology Case Study

London, England

- Proposed and presented a real-time delayed-flight dashboard with supporting data pipeline architecture to a panel of Expedia engineers, placing **2nd place**

Barclays

Apr 2026

Spring Week

Manchester, England

- Analysed fintech infrastructure and digital systems architecture through engineering workshops
- Applied structured problem-solving to real-world technology challenges in a competitive cohort setting

PROJECTS

StayR: Full-Stack Property Rental Platform | Node.js, REST API, JavaScript, Bootstrap, Git

- Built and deployed a full-stack rental platform (Node.js/Express REST API + JS/Bootstrap front-end) supporting end-to-end property search, listing management, and a bids/counter-offers negotiation flow
- Designed and implemented REST API endpoints to fetch, render, and update listings with robust error handling
- Built modular front-end logic that dynamically generates and updates DOM elements from back-end responses, improving scalability and maintainability

Traffic Signal Optimisation Pipeline | Python, OpenCV

- Built a Python pipeline using YOLOv8 and ByteTrack to extract per-second vehicle arrival rates from intersection footage
- Validated arrival data against a Poisson distribution using chi-squared testing, then optimised signal timings via particle swarm optimisation
- Ran 500-iteration Monte Carlo simulations to model uncertainty in vehicle clearance time

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML/CSS, C++

Back-end & Frameworks: Node.js, REST API design, React, Bootstrap, PostgreSQL

AI/Data: OpenAI API integration, pandas, NumPy, OpenCV, Matplotlib

Key Systems Concepts: Git, ETL pipelines, Event-Driven Architecture, Agile Development